

Vision Extract: Subject Isolation through Image Segmentation

An AI-powered approach for automatic subject isolation

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INTRODUCTION

VisionExtract is an advanced deep learning project aimed at automating the process of subject isolation from digital images.

The system employs **semantic segmentation techniques** to accurately differentiate the main subject from its background at the pixel level. This approach minimizes manual intervention and enhances the efficiency of image editing and processing workflows. The model has been deployed as an interactive **Streamlit web application**, enabling real-time image upload and subject extraction.

PROBLEM STATEMENT

Manual subject isolation from images is time-consuming, error-prone, and often dependent on expensive or limited software tools. There is a need for an automated, efficient, and accurate method to extract the main subject from any image.

Challenge:

Automatically isolating subjects from complex backgrounds requires high pixel accuracy.

Goal:

Achieve accurate subject isolation from diverse backgrounds

AI:

Leverage deep learning for automated segmentation

OBJECTIVES

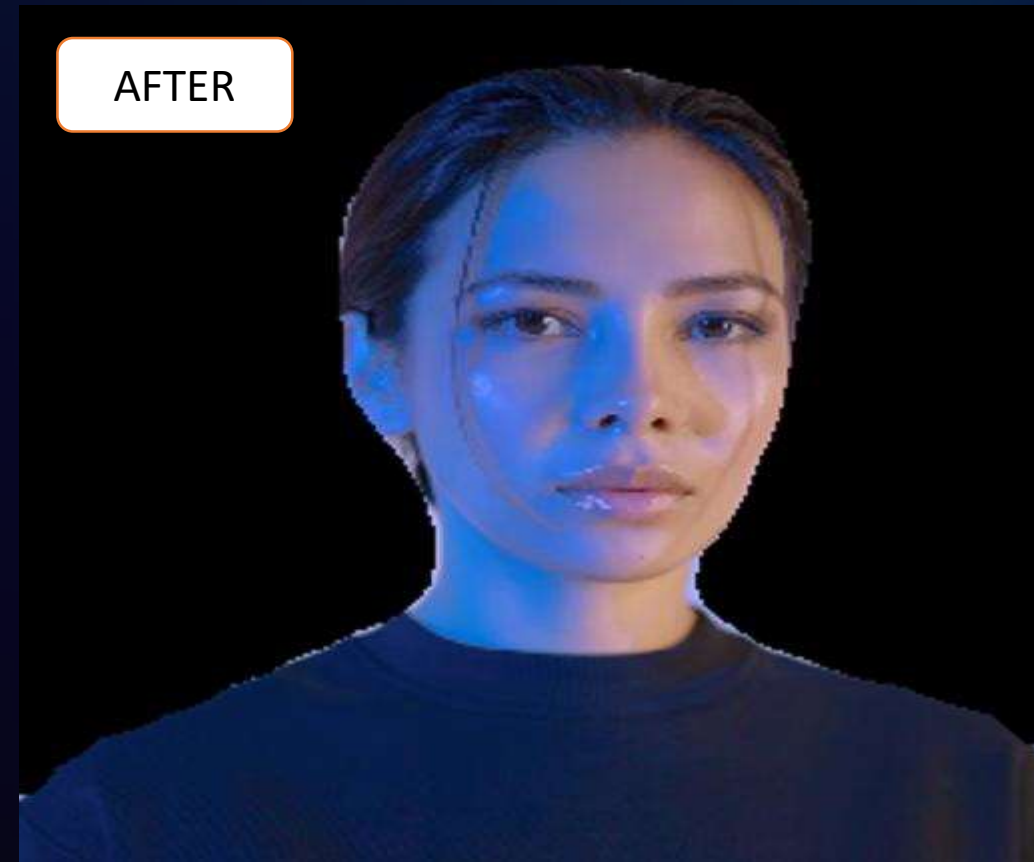
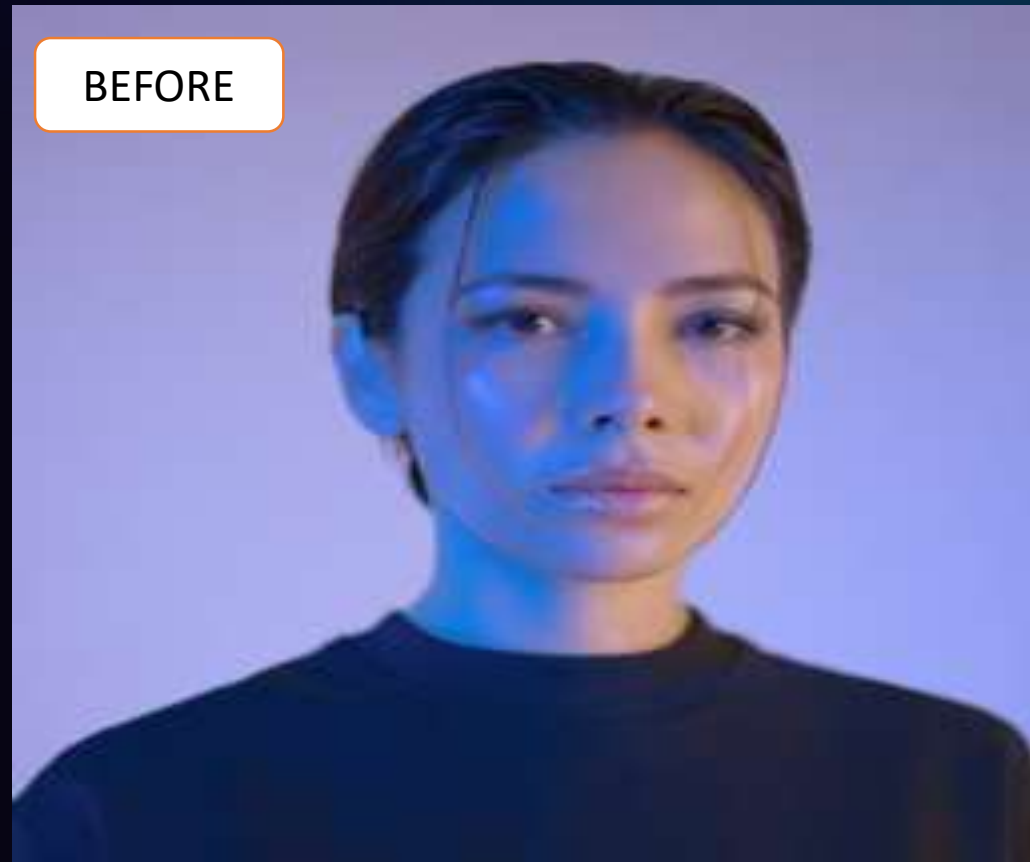
Main Objective:

- To develop an intelligent image segmentation system, **VisionExtract**, capable of automatically isolating the main subject from any image using deep learning.

Key Objectives:

- Implement **semantic segmentation** to achieve pixel-level subject extraction with high accuracy.
- Enable users to **upload images** and generate isolated subject outputs automatically.
- Provide **side-by-side visualization** of the original and masked images for better comparison.
- Allow users to **download the masked images** seamlessly.
- Deploy the complete pipeline on **Streamlit**, ensuring a clean, user-friendly, and interactive interface.
- Optimize model performance through **fine-tuning and evaluation** using metrics like IoU and Dice Coefficient.

Use Case: Automated Subject Isolation



- Instant subject extraction from any image
- High-precision mask generation
- Real-time processing capabilities

DATASET INFORMATION

COCO 2017 dataset provides comprehensive annotations for training robust segmentation models

DATASET	IMAGES	ANNOTATIONS	CLASSES	SOURCE
COCO 2017	118,287	1.5M+	80	Kaggle
Training	82,783	1.0M+	80	COCO
Validation	40,504	500K+	80	COCO

VisionExtract Development Workflow



Preprocessing

Model Training

Evaluation

Inference

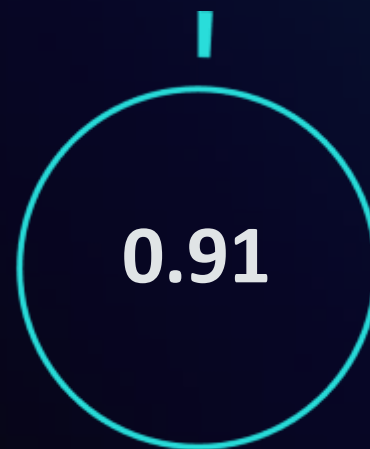
Our development journey spanned a comprehensive pipeline, ensuring robust performance from data preparation to deployment.

Evaluation: Metrics and Visual Prowess



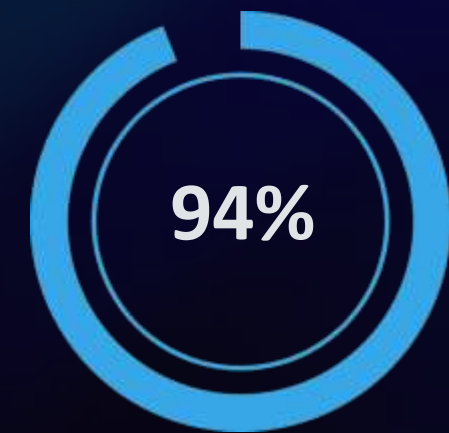
IoU Score

Intersection over Union (IoU) indicating strong overlap between predicted and ground truth masks.



Dice Coefficient

Reflecting excellent similarity and contour accuracy of the segmented subjects.

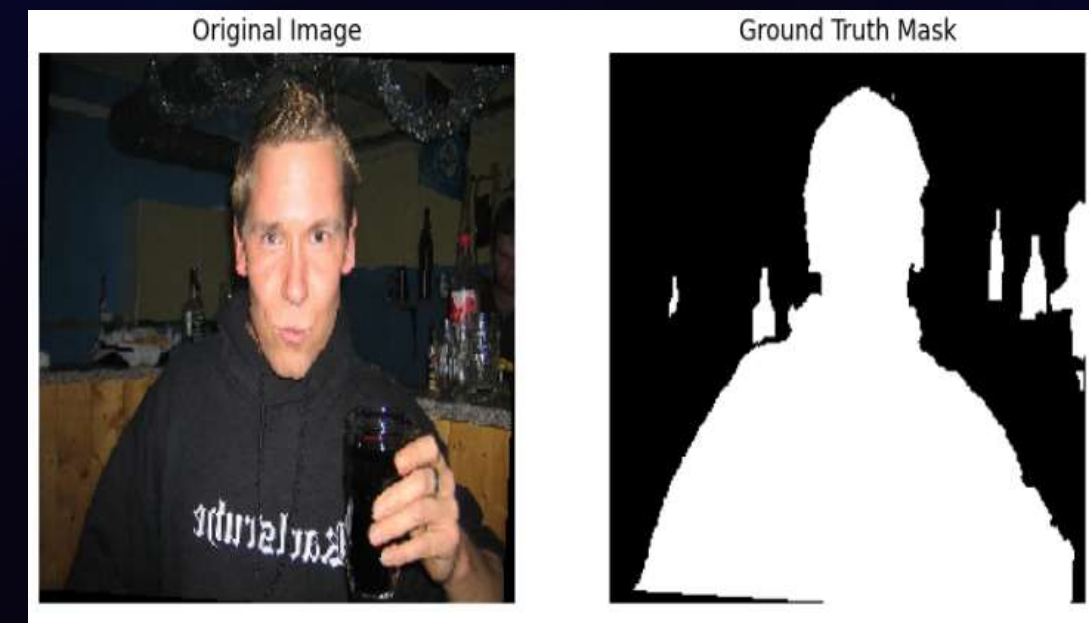
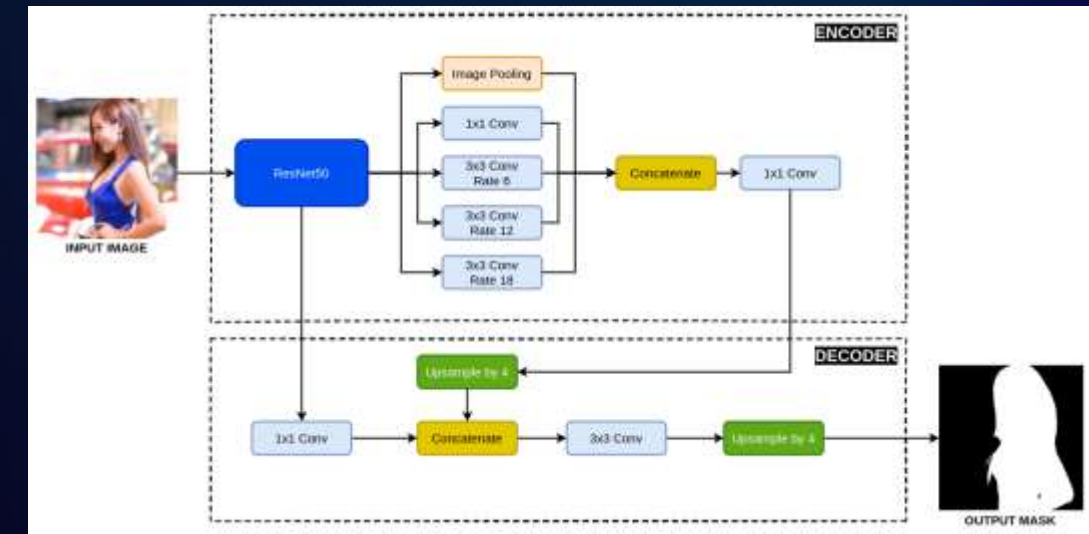


Overall Accuracy

High pixel-level accuracy in distinguishing subjects from backgrounds.

Model Architecture: DeepLabV3 with ResNet-50 Backbone

- **VisionExtract** is powered by the **DeepLabV3** architecture, a state-of-the-art deep learning model for **semantic image segmentation**.
- It utilizes a **ResNet-50 backbone** to extract high-level image features while preserving spatial details.
- The model employs **Atrous (dilated) convolutions** to capture objects at multiple scales and improve boundary precision.
- A combination of **Binary Cross-Entropy** and **Dice Loss** is used for accurate subject-background separation.
- The **output layer** uses a **Sigmoid activation** to produce a clean binary mask highlighting the isolated subject.
- DeepLabV3's **multi-scale context understanding** enables robust subject isolation even in complex backgrounds.



Overcoming Challenges: Fine-Tuning for Perfection



Hyperparameter Tuning

Optimizing learning rate, batch size, and network depth



Edge Cases

Handling complex backgrounds and overlapping objects



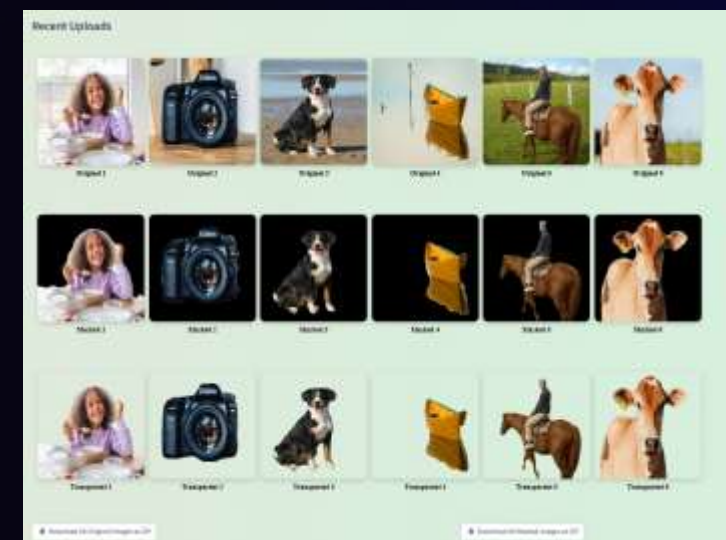
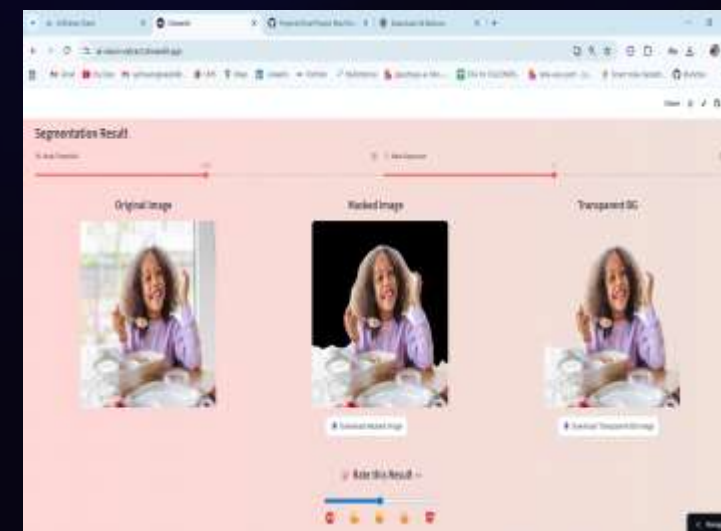
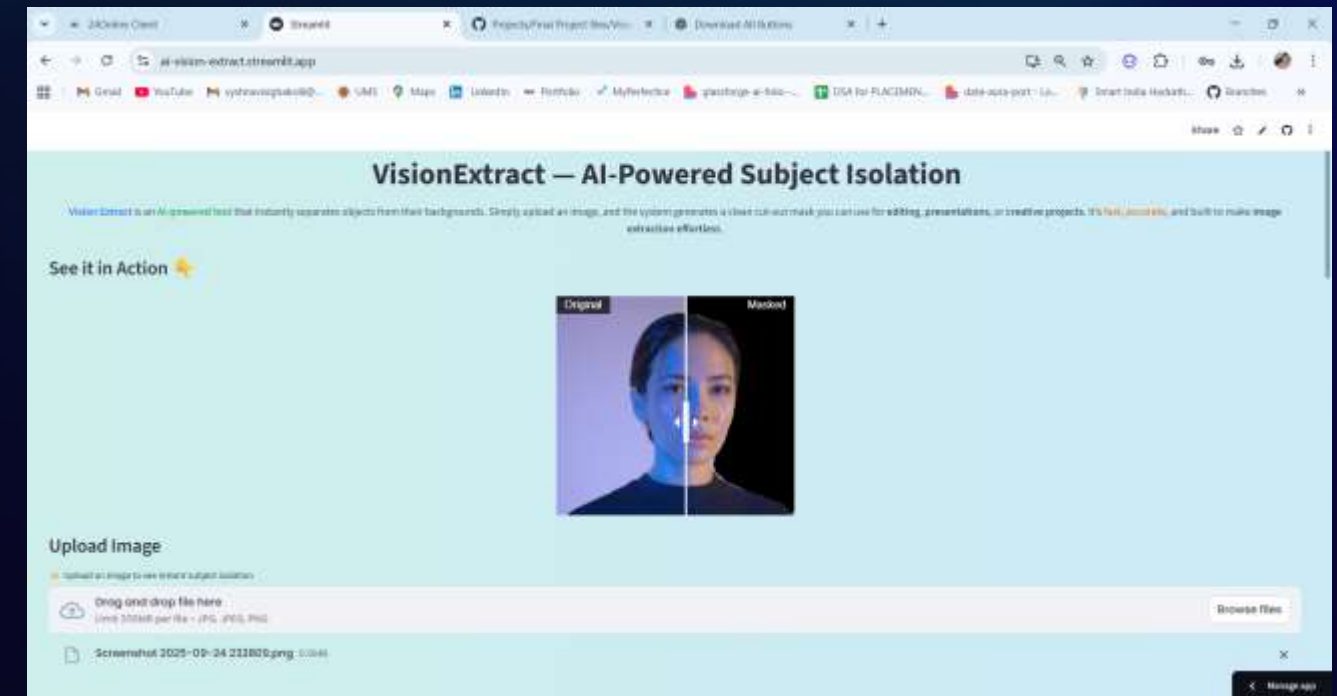
Data Augmentation

Balancing augmentation techniques to prevent overfitting

Each challenge presented an opportunity for refinement, leading to a more resilient and accurate model.

Deployment: VisionExtract Live on Streamlit

- **Instant Object Isolation:** Upload images and automatically separate subjects from backgrounds.
- **Interactive Masking:** Adjust threshold and expansion sliders to refine masks in real-time.
- **Comparison & Feedback:** Before/After comparison slider + emoji-based rating system for each image.
- **Gallery & Downloads:** View all uploads in a gallery; download individual images or all as ZIP.
- **Tech Stack:** Built with Streamlit, PyTorch, DeepLabV3, and Pillow.
- **Deployment Link:** Access the live site via [<https://ai-vision-extract.streamlit.app/>].



CONCLUSION

VisionExtract successfully automates the process of subject isolation using deep learning–based image segmentation. By leveraging the U-Net architecture, it accurately separates the main subject from complex backgrounds with minimal human effort. The project demonstrates the power of semantic segmentation in real-world applications like digital editing and AR/VR. Its Streamlit-based interface ensures an interactive and user-friendly experience for non-technical users. Overall, VisionExtract bridges the gap between advanced AI models and practical image manipulation, making segmentation accessible to everyone.

THANK YOU